

Lucas Reeves

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Education

University of California, Santa Cruz – BS in Computer Engineering	December 2025
American River College – AA in Computer Science	May 2023
American River College – AA in Math & Physical Science	May 2023

Projects

Ice Harvest – Real-Time VGA Game in Verilog github.com/charbs14/ice-harvest

- Designed and implemented a real-time, interactive VGA game on a Basys 3 FPGA using Verilog and Vivado.
- Developed modular game architecture with a finite state machine controlling a player character and 10 independent falling icicles.
- Utilized VGA protocol to render pixel-accurate graphics and a 7-segment display to track score with color-based game logic.
- Engineered collision detection, object state transitions, frame-based timing control, and flashing effects.
- Created hardware counters, synchronizers, and custom randomization logic to enhance game interactivity.
- Utilized Vivado's waveform viewer to simulate and test critical modules

IoT Environmental Monitoring System lucasreeves.me/projects/IoT-project.pdf

- Created a complete end-to-end IoT platform to collect, store, and visualize sensor data using Raspberry Pi devices in an ad-hoc network with a team of 2 other students
- Deployed sensor nodes using polling and token-ring topologies to gather temperature, humidity, wind speed, and soil moisture
- Stored sensor data in a MySQL database on a remote host using SQL insertions via Python MySQL connector, and created a Flask-based web app to visualize real time and historical data in time-series graphs.
- Developed Python scripts to simulate network topologies to test configurations

ESP32C3 Morse Code Translator github.com/charbs14/esp32c3-morse-code-translator

- Built a real-time Morse code translation system using an ESP32C3 microcontroller and its built-in ADC.
- Programmed a Raspberry Pi to blink an LED with Morse code messages via GPIO using Python.
- Interfaced a photodiode with the ESP32C3 to detect LED blinks and decode them into English characters.
- Implemented signal processing and threshold detection logic in C using the ESP-IDF framework.
- Iteratively optimized translation speed by tuning ADC sampling and increasing the device's clock frequency.

Experience

Sales Associate, Fleet Feet – Roseville, CA July 2020 – July 2023

- Delivered personalized service by listening to customer needs and recommending optimal running gear.
- Adapted quickly in a fast-paced setting by managing inventory, POS systems, and store presentations during high-traffic periods (including days, evenings, weekends, and holidays).
- Strengthened communication and collaboration by working closely with teammates and vendor partners to meet store goals and uphold Fleet Feet's "Put People First" and "Do the Right Thing" values.

Skills

Programming Languages: C, C++, Python, Verilog, SystemVerilog, JavaScript

Hardware/Embedded Systems: UART, GPIO, JTAG, I2C, SPI, Oscilloscopes

Web/Database Technologies: Flask, MySQL, MariaDB, HTML/CSS (basic)

Tools & Platforms: Vivado, ESP-IDF, Linux (Ubuntu), Windows, Git, VS Code